Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/08/2011
PI Name:	Sams, Clarence Ph.D.		
Project Title:	Immune Function Changes During a Spaceflight-analog 12-day Undersea Mission (NEEMO IIH ROI)		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	clarence.sams-1@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281-483-7160
Organization Name:	NASA Johnson Space Center		
PI Address 1:	Human Adaptation and Countermeasures Office		
PI Address 2:	2101 NASA Parkway, Mail Code SK		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3607	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	05/01/2007	**End Date:**	03/25/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Baumann, David	**Contact Phone:**	
Contact Email:	david.k.baumann@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/25/2011 (from 10.31/2009) per discussion with CoI (Ed., 8/10/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crucian, Brian (NASA Johnson Space Center) Stowe, Raymond (Microgen) Pierson, Duane (NASA Johnson Space Center) Mehta, Satish (NASA Johnson Space Center) Chouker, Alexander (Hospital of the Ludwig-Maximilians-University, Munich, Germany)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Spaceflight-associated immune dysregulation occurs during spaceflight and may represent specific clinical risks for exploration-class missions. An appropriate ground analog for spaceflight-associated immune dysregulation (SAID) would offer a platform for ground evaluation of various potential countermeasures. This study evaluated the NASA Extreme Environment Mission Operations (NEEMO), consisting of 14 day undersea deployment at the Aquarius station, as an analog for SAID. Some important potential causes for in-flight immune dysregulation are shared with space analogs such as NEEMO: confinement, isolation, health risk (EVA, saturation diving), physiological stress, disrupted circadian rhythms. Specific assays included peripheral immunocyte distribution, T cell function, cytokine profiles, virus-specific immunity, and latent herpesvirus reactivation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study has assessed immune system, stress, and viral reactivation alterations in a high-fidelity ground-based spaceflight analog. Missions consist of undersea confinement, saturation diving, significant EVA activity, and associated health risks. The data confirms physiological changes similar to those observed during spaceflight. These data should be applicable to other similar terrestrial situations, such as undersea naval missions or government/scientific deployment to extreme environment stations (arctic, Antarctic, etc.). Thus, the monitoring strategy developed by this study, and any future countermeasures development, should have terrestrial benefit for these analogous situations.
Task Progress:	This task is concluded. All experimental objectives were successfully completed on three NEEMO missions: NEEMO-12 (May 7-18, 2007); NEEMO-13: (August 6-15, 2007); NEEMO-14: (May 10-23, 2010). A total of sixteen NASA 'Aquanauts' participated in this study. A summation of the study results follows: Mid-mission alterations leukocyte distribution occurred, including granulocytosis and elevations in central-memory CD8+ T-cells. General T cell function was reduced during NEEMO missions in roughly 50% of subjects. Secreted cytokines profiles were evaluated following whole blood stimulation with CD3/CD28 (T cells) or LPS (monocytes). T cell production of IFNg, IL-5, IL-10, IL-2, TNFa and IL-6 were all reduced before and during the mission. Conversely, monocyte production of TNFa, IL-10, IL-6, IL-1b and IL-8 were elevated during mission, more so at the MD-14 time point. Granulocyte adhesion molecule expression (e.g. CD11b, CD62L) as assessed in NEEMO-14 indicated high activation during mission. Antibodies to Epstein-Barr virus (EBV) viral capsid antigen and early antigen were increased in approximately 40% of the subjects. Changes in EBV tetramer-positive CD8+ T-cells exhibited a variable pattern. Antibodies against Cytomegalovirus (CMV) were marginally increased during the mission. Herpesvirus reactivation was determined by PCR. EBV viral load was generally elevated at L-6. Higher levels of EBV reactivation were found during the NEEMO missions, however no VZV or CMV reactivation was observed in any of the NEEMO mission or control samples. Plasma cortisol was elevated at L-6. Summation of data: some changes in leukocyte distribution, T cell function, cytokine production, virus specific immunity and viral reactivation that occur during NEEMO missions are similar to those observed during or immediately following spaceflight. Unfortunately, 6 days prior may be too near to mission start to serve as an appropriate baseline measurement. The NEEMO platform may have utility for short-duration, ground-based spaceflight-immune research, such as investigations of mechanism or countermeasures validation.
Bibliography Type:	Description: (Last Updated: 07/09/2025)
Abstracts for Journals and Proceedings	Crucian BE, Stowe RP, Mehta SK, Feuerecker M, Choukèr A, Quiriarte H, Pierson DL, Sams CF. "NASA 14 Day Undersea Missions: A Short-Duration Spaceflight Analog for Immune System Dysregulation." 18th IAA Humans in Space Symposium, Houston, Texas, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, Texas, April 11-15, 2011. , Apr-2011
Articles in Peer-reviewed Journals	Strewe C, Crucian BE, Sams CF, Feuerecker B, Stowe RP, Choukèr A, Feuerecker M. "Hyperbaric hyperoxia alters innate immune functional properties during NASA Extreme Environment Mission Operation (NEEMO)." Brain Behav Immun. 2015 Nov;50:52-7. Epub 2015 Jun 25. https://doi.org/10.1016/j.bbi.2015.06.017 ; PubMed PMID: 26116982 , Nov-2015